

MAT 1013: Discrete Mathematics

Session 2: Truth Tables and Logical Equivalence

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Learning Outcomes

By the end of this session, students should be able to:

- **Construct** truth tables for logical expressions
- **Identify** tautologies and contradictions
- **Determine** logical equivalence
- **Compare** symbolic and tabular reasoning

Part 1: Truth Values of Compound Propositions

Activity

Evaluate:

- | | |
|-------------------------------------|--|
| 1. $\neg p$ when $p = T$ | 4. $p \rightarrow q$ when $p = F, q = F$ |
| 2. $p \wedge q$ when $p = T, q = F$ | 5. $p \rightarrow q$ when $p = T, q = F$ |
| 3. $p \vee q$ when $p = F, q = F$ | 6. $\neg(p \vee q)$ when $p = F, q = F$ |

Discussion: What determines truth values?

Part 2: Truth Tables

A truth table is a table that lists all possible combinations of truth values for the propositional variables in a logical expression, and, for each combination, shows the resulting truth value of the entire expression.

Questions:

1. Let p, q and r are propositional variables.
 - (a) then list all possible truth value combinations of p, q and r .
 - (b) Construct the truth table for $p \wedge (q \wedge r)$.
2. Construct the truth table for $\neg(\neg p)$, where p is a propositional variable.
3. If there are n propositional variables then how many rows are there in the truth table?
I.e., How many T/F combinations are there?

Examples

Complete the truth tables for $p \wedge q$ and $p \vee q$.

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Part 3: Truth tables of standard logical operations

Conjunction

$p \wedge q$		
p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

- $p \wedge q$ is **true** exactly when both p and q are true.
- It is **false** if at least one of p or q **false**.

Disjunction

$p \vee q$		
p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

- $p \vee q$ is **false** exactly when both p and q are **false**.
- It is true if at least one of p or q **true**.

Negation

$\neg p$	
p	$\neg p$
T	F
F	T

- $\neg p$ is true if p is **false**.
- $\neg p$ is false if p is **true**.

Implication

$p \implies q$		
p	q	$p \implies q$
T	T	T
T	F	F
F	T	T
F	F	T

- $p \implies q$ is **false** exactly when p is **true** and q is **false**.
- It is **true** whenever the premises p is false or when p is **true** and the consequent q is **false**.

Biconditional (if and only if)

$p \iff q$		
p	q	$p \iff q$
T	T	T
T	F	F
F	T	F
F	F	T

- $p \iff q$ is **true** exactly when p and q take the same truth value (i.e., both true or both false).
- It is **false** whenever p and q have different truth values.

Questions:

1. construct the truth table for the following propositions. What could you conclude from these truth tables.
 - (a) $p \vee (\neg q)$
 - (b) $(\neg q) \implies (\neg p)$
2. Construct the truth table for the following propositions.
 - (a) $(p \vee q) \wedge \neg p$
 - (b) $(p \vee q) \implies r$
3. construct the truth table for the following propositions. What could you conclude from these truth tables.
 - (a) $p \vee \neg p$
 - (b) $p \implies p$
 - (c) $\neg(p \wedge \neg p)$
 - (d) $(p \wedge \neg p) \implies q$
4. Can you give examples of propositions that is always **false**.

Part 4: Tautology and Contradiction

Tautology: A proposition that is Always true (Notation : **T**)

Contradiction: A proposition that is always false (Notation : **F**)

Examples:

$$p \vee \neg p \quad (\text{Tautology})$$

$$p \wedge \neg p \quad (\text{Contradiction})$$

Part 5: Logical Equivalence

Two compound propositions are said to be logically equivalent if they have the same truth value for every possible assignment of truth values to their component propositions; that is, they have identical truth tables.

Examples: Verify the equivalence of the following using truth tables.

$$\neg(p \wedge q) \equiv \neg p \vee \neg q$$

Questions:

5. Show that the following are logically equivalent. Here, p and q are propositional variables.

- (a) $p \implies q$
- (b) $\neg q \implies \neg p$ (contrapositive)
- (c) $(\neg p) \vee q$
- (d) $\neg(p \wedge (\neg q))$.

6. Show that $\neg(\neg p)$ is logically equivalent to p .

7. Show that $p \iff q$ is logically equivalent to $(p \implies q) \wedge (q \implies p)$.

8. Construct truth table for $p \implies (q \vee p)$. Is it a tautology?

9. Write down as many logical equivalences as possible.

Important Facts

Mistake	Correction
Number of rows	Use 2^n rule
Logical OR	Inclusive OR
Materialistic implication	Only false in one case

Independent Study

- Construct tables for 3-variable expressions
- Explore patterns in implication
- Investigate De Morgan's Laws
- Investigate the distributive law

Puzzle

Four friends—Gregory, April, August, and June—are accused of breaking a window. When questioned:

- Gregory: "June did it."
- April: "August did not do it."
- August: "April did not do it."
- June: "Gregory lied when he said I did it."

If *only one person is telling the truth*, who broke the window?